Consensus-based Decentralized Distributed Swarm Learning with Heterogeneous Big Data

Zhuoyu Yao[†], Dong Yang[†], Yue Wang[†], Songyang Zhang[*] Yingshu Li[†], Zhi Tian[⋆], Zhipeng Cai[†]
[†]Department of Computer Science, Georgia State University, Atlanta, GA, USA
[*]Department of Electrical and Computer Engineering, University of Louisiana at Lafayette, LA, USA
[⋆]Department of Electrical and Computer Engineering, George Mason University, Fairfax, VA, USA

***Abstract*—Modern artificial intelligence increasingly relies on large-scale, distributed, and heterogeneous data collected by edge devices. However, the practice of edge intelligence remains challenging due to non-convex objectives, data heterogeneity, and privacy concerns. To address these issues, this paper proposes a consensus-based decentralized distributed swarm learning (CD-DSL) framework for wireless edge networks. CD-DSL integrates consensus optimization with particle swarm optimization (PSO), by leveraging the adaptive neighbor-mixing together with PSO exploration and exploitation. The consensus mechanism supports decentralized coordination without raw-data exchange, while PSO-inspired updates utilize historical and neighbor-shared experience to enhance exploration for non-convex optimization, improve robustness to data heterogeneity, and accelerate convergence. We further develop an adaptive neighbor-mixing strategy that learns performance-aware consensus weights, improving decentralized collaboration among heterogeneous edge devices. Theoretical analysis establishes that CD-DSL maintains participant consistency and achieves non-ergodic convergence to a neighborhood of a stationary point under non-convex objectives. Experimental results show that CD-DSL can mitigate the performance degradation of existing decentralized baselines caused by heterogeneous data.**

***Index Terms*—Heterogeneous big data, consensus optimization, decentralized edge network, distributed swarm learning, non-convex optimization, convergence analysis.**

I. Introduction

The rapid growth of edge intelligence in wireless networks is shifting learning and inference from centralized platforms to large-scale edge devices, enabling low-latency and real-time intelligent services [1]. However, big data generated by drones, sensors, vehicles, and mobile terminals are inherently large-scale and highly heterogeneous, rendering centralized processing impractical due to the prohibitive overhead for data transmission and computation. Distributed learning avoids raw data collection [2], but suffers from performance degradation under non-IID data and non-convex objectives [3].

Federated learning (FL) has been widely investigated to address data heterogeneity in distributed settings [4]. Unfortunately, most pioneering FL frameworks hinge on a central server for model aggregation and coordination, which is vulnerable to node/link failures and faces communication bottlenecks as well as concerns regarding trust in the central server [5]. These limitations motivate a more robust and efficient decentralization for edge intelligence. However, replacing the central server with peer-to-peer collaboration does not automatically produce effective decentralized learning. Without a global coordinator, decentralized learning suffers from slower convergence and weaker agreement guarantees, especially with massive devices with heterogeneous data.

Consensus optimization provides a principled paradigm for coordination and agreement among decentralized devices [6], where consensus mechanisms enable neighboring devices to iteratively exchange model information and reach agreement without relying on a central server. However, classical consensus formulations are mainly developed for convex optimization settings and often assume doubly stochastic weights, which does not fit practical settings. In real-world edge networks, heterogeneous device capabilities, asymmetric communication links, and dynamic wireless conditions frequently lead to directed and imbalanced information flows [7]. Thus, conventional consensus-based learning methods may suffer from biased aggregation, degraded convergence, and limited robustness under realistic wireless network conditions. While decentralized learning under non-convex objectives has been studied in [7]–[9], theoretical analysis remains unattended.

Even worse, data and system heterogeneity are intrinsic characteristics of wireless edge networks, where heterogeneity becomes more challenging in decentralized settings because each device only has a limited local perspective and cannot rely on a central server to provide a common evaluation criterion. Existing decentralized stochastic gradient descent (DSGD) methods primarily focus on improving local update rules or communication efficiency [10], but not aiming at promoting collaboration among heterogeneous participants [11]. As a result, when local data distributions are highly non-IID, different devices may produce conflicting model updates, resulting in slow convergence, poor generalization, and vulnerability to local traps in non-convex cases.

Our previous works on distributed swarm learning (DSL) suggest a complementary direction to improve efficiency, robustness, and convergence [12]–[14]. By integrating AI with biologically inspired swarm intelligence, DSL leverages local historical experience and collaborators' information to improve learning efficiency under heterogeneous data distributions [15]. However, DSL cannot be straightforwardly applied in decentralized wireless networks as the existing designs often depend on a shared evaluation perspective to score

This work was supported in part by the NSF grants #2146497, #2231209, #2244219, #2315596, #2343619, #2349878, #2416872, and #2413622.

particles or identify the best-performing model, which is not readily available in serverless decentralized systems. Therefore, standard DSL lacks a principled aggregation mechanism for weighting neighbors over communication graphs.

To address the above challenges, this paper proposes a consensus-based decentralized distributed swarm learning (CD-DSL) framework for robust and efficient edge intelligence systems with heterogeneous big data. CD-DSL extends distributed learning to fully decentralized edge systems by integrating consensus optimization with swarm learning, achieving faster convergence, efficient exploration and enhanced exploitation. We further develop an adaptive neighbor-mixing strategy to improve exploitation and collaboration under heterogeneous input data. Meanwhile, the swarm-intelligence component enables exploration to escape local optima for non-convex scenarios. Our contributions are listed as follows.

1) In CD-DSL, we design a bio-inspired decentralized learning paradigm, which addresses non-IID big data, privacy concerns, and non-convex issues by integrating consensus-based decentralized learning with DSL exploration-and-exploitation without global evaluation.
2) We develop an adaptive neighbor-mixing strategy to learn adaptive consensus weights from local validation feedback. This mechanism enables each device to evaluate neighbors' models locally and assign higher weights to more beneficial collaborators, improving decentralized learning with heterogeneous input.
3) We incorporate a push-sum consensus mechanism to correct aggregation bias caused by stochastic mixing over directed graphs. This allows CD-DSL to operate in realistic asymmetric edge networks instead of relying on impractical doubly stochastic assumptions.
4) Theoretical analysis indicates that CD-DSL encourages participant consensus and achieves non-ergodic convergence to a neighborhood of a stationary point under non-convex objectives, and characterizes how data heterogeneity and hyperparameters affect convergence.

Notations: Let $\mathbf{1} \in \mathbb{R}^{n\times 1}$ be an all-one column vector and $\mathbf{I} \in \mathbb{R}^{n\times n}$ represent an identity matrix. We apply an event indicator $\mathbb{I}_{(f_{i,t-1}>f_{i,t})}$, whose expectation over time t describes the probability of $f_{i,t-1} > f_{i,t}$. Given a matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\|\mathbf{A}\|$ is the spectral norm. And $\|\mathbf{x}\|$ denotes the Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^{n\times 1}$. Hadamard product and division of two matrices are expressed as $\mathbf{A} \circ \mathbf{B}$ and $\mathbf{A} \oslash \mathbf{B}$.

II. CD-DSL: Consensus-based Decentralized Distributed Swarm Learning

This section formulates the decentralized learning problem, followed by a consensus mechanism design for directed topologies [8], which leads to our CD-DSL algorithm design.

A. Problem Formulation

Consider a decentralized wireless network with K devices collaboratively performing a shared task. Its topology can be represented as a strongly connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with the set of devices $\mathcal{V}$ and that of wireless communication links $\mathcal{E}$. The adjacency matrix $\mathbf{A} = \{a_{i,j}\}_{K\times K}$ characterizes the peer-to-peer communication accessibility, in which $a_{i,j} = 1$ denotes a reliable communication link from device i to device j. Devices can only reach out their one-hop neighbors for sharing model-training local updates (raw data query is prohibited). We set $a_{i,i} = 1, \forall i$ for self-accumulated training gains from local historical experiences. Each participant maintains a local learning model $M_i(\mathbf{w}_i, D_i), \mathbf{w}_i \in \mathbb{R}^{N\times 1}, i = 1, \ldots, K$ with parameter vector $\mathbf{w}_i$ and local observation dataset D_i. It is worth noting that: caused by the biased local observations and heterogeneous data distributions, standalone training suffers from suboptimal global solutions, due to the lack of collaboration among decentralized devices. It motivates establishment of a global objective of edge-intelligence systems, aiming a general model $\overline{\mathbf{w}}$ to mitigate heterogeneity impact:

$$\overline{\mathbf{w}} \triangleq \arg\min_{\mathbf{w}} \frac{1}{K}\sum_{i=1}^{K} f_i(\mathbf{w}, D_i), \tag{1}$$

where loss function f_i is evaluated based on local dataset D_i. Local model updates follow a weighted local decentralized gradient descent mechanism for the generalization of $\overline{\mathbf{w}}$ [16]:

$$\mathbf{w}_{i,t+1} = \sum_{k\in\mathcal{N}_i\cup\{i\}} p_{k,i}\mathbf{w}_{k,t} - \alpha_i \nabla f(\mathbf{w}_{i,t}, D_i), \forall i, \tag{2}$$

where α_i denotes the learning rate of i-th model, $p_{k,i}$ is a trainable parameter representing the alignment of neighboring devices k to the device i. $\mathcal{N}_i \cup \{i\}$ represents the neighborhood of device i, where $\mathcal{N}_i = \mathcal{N}_i^{in} \cup \mathcal{N}_i^{out}$ includes both in-degree and out-degree neighbors. For directed networks, $\mathcal{N}_i^{in}$ denotes the set of neighbors from whom device i can reliably receive messages, whereas $\mathcal{N}_i^{out}$ includes the neighbors to whom device i can transmit its local updates. In this sense, each client can share own historical experience and benefit from the neighbor's current training updates. A state space model is defined for the global iteration of Markov training process $\mathbf{W}_{t+1} = \mathbf{W}_t \cdot \mathbf{P}_{K\times K} - \Lambda, i = 1, \ldots, K$, where $\mathbf{P} = [\mathbf{P}_i] = \{p_{i,j}\}$ is the mixing matrix to aggregate updates from neighbors. $\mathbf{P}$ is initialized by the column normalizing $\mathbf{A}$ with $p_{i,j}^t \equiv 0, \forall t$ for all unconnected node pairs. Thus, $\sum_{k\in\mathcal{N}_i\cup\{i\}} p_{k,i} = \sum_{k=1}^{k=K} p_{k,i}, \forall i$ and we enlarge the integration interval from devices' neighborhood to the complete set for concise expression. $\Lambda \triangleq (\alpha_i \nabla f(\mathbf{w}_{i,t}, D_i))_1^K \in \mathbb{R}^{N\times K}$ denotes a stacked matrix with the local gradient, where $f(\mathbf{w}_{i,t}, D_i)$ can be denoted as $f_{i,t}$.

B. Push Sum Consensus

Consensus optimization orchestrates coordination for consistency of distributed local training through consensus-based constraints to encourage cooperation across participants [17]. Given (2), the problem (1) in decentralized networks becomes:

$$\begin{aligned} \arg\min_{\mathbf{W}} \frac{1}{K}\mathbf{1}^\top \cdot \mathbf{f}(\mathbf{W}, \mathbf{D}) &\triangleq \frac{1}{K}\sum_{i=1}^{K} f_i(\mathbf{w}_i, D_i) \\ s.t. \quad \mathbf{w}_i &= \mathbf{w}_j, \forall (i,j) \in \mathcal{E}, \end{aligned} \tag{3}$$

where the vector $\mathbf{f}(\mathbf{W},\mathbf{D}) \in \mathbb{R}^{1\times K}$ records the agents' loss functions and gradients of model parameters $\mathbf{W} \in \mathbb{R}^{N\times K}$.

The mixing matrix $\mathbf{P}$ performs the consensus-based aggregation scheme. To guarantee convergence, an assumption is usually applied to hold a doubly stochastic $\mathbf{P}$ in order to aggregate updates from neighbors within undirected wireless topologies [18]. However, the data and devices' heterogeneity in edge networks leads to asymmetric interactions, violating such an assumption of the symmetric and doubly stochastic property of $\mathbf{P}$. To fill this gap, we next investigate a robust consensus scheme for directed topologies, which is reliable to apply for heterogeneous data in wireless edge networks.

Assumption 1. *(Column stochastic mixing matrix): Given a strongly connected network topologies $\mathcal{G}$, there exists an optimal mixing matrix $\mathbf{P}^\star = \{p^\star_{i,j}\} \in \mathbb{R}^{K\times K}$ is nonnegative, off-diagonal entries and column stochastic as:*

$$\mathbf{1}^\top \mathbf{P} = \mathbf{1}; \quad 0 < p_{i,j}, \forall (i,j) \in \mathcal{E}; \quad 0 \le p_{i,j} \le 1. \tag{4}$$

The eigenvalues of $\mathbf{P}$ follow $1 = \lambda_1(\mathbf{P}) \ge |\lambda_2(\mathbf{P})| \ge \ldots \ge |\lambda_K(\mathbf{P})|$.

Apart from the existing doubly stochastic mixing matrix design [8], [19], $\mathbf{P}$ forms a column stochastic, nonnegative and off-diagonal paradigm. The relaxed condition of **Assumption 1** than that of [9], not only preserves its validity, but also makes it more suitable for directed and heterogeneous wireless edge networks. The consensus optimization allows to find a universal model to mimic the global optimal model such that all trained models will converge to the same neighborhood of given stationary point $\mathbf{w}_1, \ldots, \mathbf{w}_K, \overline{\mathbf{w}} \in [\mathbf{w}^\star - \epsilon, \mathbf{w}^\star + \epsilon]$.

Although the column stochastic aggregation graph reduces modeling complexity and executes better in directed topological structure, the substituted row stochastic incurs an imperative gap as $1/K \sum_i \mathbf{w}_{i,\infty} \neq 1/K \sum_j \mathbf{w}_{j,\infty}$ as $\sum_{i=1}^{i=K} p_{i,j} \neq 1$. This gap incurs a biased objective $\overline{\mathbf{w}} = \arg\min_{\mathbf{w}} 1/K \sum_i \mathrm{Pr}_i f_i(\mathbf{w}_i, D_i)$ from the optimal aggregation, where the coefficient Pr_i is attributed to the deviant stationary distribution of $\mathbf{P}$. Inspired by the push-sum consensus mechanism [8], we introduce a nonnegative scalar matrix $\mathbf{Z} = \{\mathbf{z}_i\}_1^K, \mathbf{z} \in \mathbb{R}^{N\times 1}$ to rebuild symmetric matrix for global updating. With the initialized $\mathbf{z}_{i,0} = \mathbf{1}, \forall i$, individual parameter updates follow a ratio consensus prototype:

$$\begin{aligned}
\mathbf{W}_{t+1} &= (\mathbf{W}_t \cdot \mathbf{P}) \oslash (\mathbf{Z}_t \cdot \mathbf{P}) - \Lambda \\
\mathbf{Z}_{t+1} &= \mathbf{Z}_t \cdot \mathbf{P} = \mathbf{Z}_0 \cdot \mathbf{P} \\
\widetilde{\mathbf{W}}_{t+1} &= (\mathbf{W}_{t+1} \circ \mathbf{Z}_{t+1})^{\circ -1},
\end{aligned} \tag{5}$$

where $\mathbf{Z}$ reveals the accumulated aggregated gains on the scalar matrix. $\widetilde{\mathbf{W}}$ is the ratio of parameter $\mathbf{w}_{j,t}$ and scalar $\mathbf{z}_{j,t}$ for unbiased aggregation. Devices transmit $\widetilde{\mathbf{W}}$ as an adjusted individual gain to their neighbors. The receiving devices sum up all the information with column stochastic $\sum_{j=1}^{j=K} p_{j,i} = 1$ to promote consistency. Based on Perron-Frobenius [20] validation $\lim_{t\to\infty} \mathbf{W}_t \circ \mathbf{Z}_t^{\circ -1} = 1/K \mathbf{1}\mathbf{1}^\top \mathbf{W}_0$, consensus algorithm $\overline{\mathbf{w}}_{t+1} = 1/K \sum_{i=1}^{i=K} (\widetilde{\mathbf{w}}_{i,t} - \alpha_i \nabla f(\widetilde{\mathbf{w}}_{i,t}))$ will converge to a specific neighborhood with proper learning rates and sufficient iteration. Push sum mechanism guarantees the consensus recognition at the deployment of decentralized learning frameworks in directed wireless topologies. Without the row stochastic restrictions, the consensus scheme benefits to learn a more flexible $\mathbf{P}$ for the precise and reliable approximation about topological interactions.

Although such a mechanism guarantees the generalization and non-ergodic convergence of decentralized learning paradigm for edge computing, data heterogeneity and non-convex objectives in wireless networks still prevent consensus based decentralized learning algorithm from robust and efficient implementation.

C. CD-DSL Algorithm

The prevalence of non-IID observations, non-convex downstream tasks, and one-hop communication constraints calls for an urgent demand for robust decentralized learning framework in wireless edge networks. Edge intelligence enables an effective way to improve the consistency and generalization of distributed learning under heterogeneous inputs. Adaptive model aggregation can alleviate slow convergence by enriching local decisions with valuable experience from neighboring agents and strengthening exploration for improved modeling simultaneously [7]. Meanwhile distributed swarm learning (DSL) has demonstrated strong robustness to heterogeneity and enhanced exploration and exploitation in non-convex optimization, making it a natural complement to consensus-based decentralized learning. The learning scheme of vanilla DSL implements parameter updates as [12], [13]:

$$\begin{aligned}
\mathbf{v}_{i,t+1} =& \mathbf{w}_{i,t+1} - \mathbf{w}_{i,t} \\
=& c_0 \mathbf{v}_{i,t} + c_1 \left(\mathbf{w}^l_{i.t} - \mathbf{w}_{i,t}\right) + c_2 \left(\mathbf{w}^g_t - \mathbf{w}_{i,t}\right) \\
& - \alpha \nabla F\left(\mathbf{w}_{i,t}, D_g\right),
\end{aligned} \tag{6}$$

where the velocity $\mathbf{v}_{i,t}$ is the increment of model parameters in device i at t communication round, $\mathbf{w}^l_{i,t}$ and $\mathbf{w}^g_t$ denotes the local and global best model for the exploitation of outstanding experience. The inertia hyperparameter c_0 controls the exploration capability of model updating. The coefficients c_1, c_2 regulate the exploitation of local and global experience. The exploration term $c_0 \mathbf{v}_{i,t}$ enables the local model to perform exploratory updates that may temporarily deviate from locally optimal directions, strengthening the ability to escape saddle points and search for globally favorable solutions.

The local best $\mathbf{w}^l_{i.t}$ and global best design $\mathbf{w}^g_t$ serve as the indispensable guidance, enabling each agent to exploit both its historical best solution and valuable experience collected from neighboring participants. Such enhanced network intelligence improves the update efficiency, model generalization and collaborative scheme, in which the PSO-based update mechanism mitigates conflicting training gains with heterogeneous input.

However, in the decentralized networks, the one-hop communication constraints limit the global horizon of local devices, impeding participants gain immediate update information from those out of their neighborhoods. Facing this issue, to apply DSL in a decentralized manner, we intro-

duce a consensus oriented neighbor best model to record the outstanding training gains from one hop neighbors. The flooding-based information dissemination mechanism allows individual devices in a decentralized network to progressively obtain a global view through iterative neighbor-to-neighbor information exchanges [21]. Thanks to the long-term accumulation of neighbor-shared experience, the neighbor-best model gradually serves as an effective approximation to the global-best guide, leading the local models toward the global optimum. Based on the weighted parameter matrix $\widetilde{\mathbf{W}}$, the exploitation term is formulated as

$$\begin{aligned}\widetilde{\mathbf{W}}_t^l =& \left[\mathbb{I}_{(f_{i,t}>f_{i,t-1})}\right]_{i=1}^K \cdot \mathbf{1} \circ \widetilde{\mathbf{W}}_{t-1} \\ &+ \left[\mathbb{I}_{(f_{i,t-1}>f_{i,t})}\right]_{i=1}^K \cdot \mathbf{1} \circ \widetilde{\mathbf{W}}_t \qquad (7)\\ \widetilde{\mathbf{W}}_t^n =& \widetilde{\mathbf{W}}_t \cdot \mathbf{P}, \qquad (8)\end{aligned}$$

where the event indicator $b_{i,t} \triangleq \mathbb{I}_{(f_{i,t-1}>f_{i,t})}$ denotes the probability that recent model on device i performs better than previous one. As the training process follows a Markov chain, $\widetilde{\mathbf{W}}_t^l$ records the historical best parameters. Then the consensus-based decentralized DSL performs model updates:

$$\begin{aligned}\widetilde{\mathbf{W}}_{t+1} =& \widetilde{\mathbf{W}}_t \cdot \mathbf{P} - \Lambda(\widetilde{\mathbf{W}}_t) + c_0 \mathbf{V}_t \qquad (9)\\ &+ c_1 \mathbf{B}_t \circ \left(\widetilde{\mathbf{W}}_{t-1} - \widetilde{\mathbf{W}}_t\right) + c_2 \widetilde{\mathbf{W}}_t \cdot (\mathbf{P} - \mathbf{I}),\end{aligned}$$

where the probability vector $\mathbf{B} = \left[\mathbb{I}_{(f_{i,t-1}>f_{i,t})}\right] \cdot \mathbf{1} \in \mathbb{R}^{K\times K}$ represents a binary matrix to formulate accumulated training gains, $\mathbf{V}_t = [\mathbf{v}_{i,t}]_{i=1}^K$ records the exploration velocity of participants to escape from local trap. $\mathbf{I}$ represents the identity matrix. For each local updates:

$$\begin{aligned}\widetilde{\mathbf{w}}_{i,t+1} =& (1+c_2)\sum_j p_{j,i}\widetilde{\mathbf{w}}_{j,t} - \alpha_i \nabla f_{i,t} + c_0 \mathbf{v}_{i,t} \qquad (10)\\ &+ c_1 \mathbb{I}_{(f_{i,t-1}>f_{i,t})}\left(\widetilde{\mathbf{w}}_{i,t-1} - \widetilde{\mathbf{w}}_{i,t}\right) - c_2 \widetilde{\mathbf{w}}_{i,t}, \forall i.\end{aligned}$$

By incorporating the neighbor best model $\widetilde{\mathbf{w}}_t^n$, the mixing matrix serves not only to enforce consensus among local updates, but also to realize adaptive neighborhood aggregation. Accordingly, we design a joint optimization formulation to reflect the wireless communication topology and adaptive model comparison under heterogeneous inputs. At current communication round t, client i learns an adaptive incoming consensus weight vector $\mathbf{p}_i^t = \{p_{j,i}^t\}_{j=1}^K$ by evaluating neighbor models on its local validation data D_i^{val}. The trainable consensus weights are obtained by solving

$$\begin{aligned}\mathbf{p}_i^t = \arg\min_{\mathbf{p}_i} & \sum_{j\in\mathcal{N}_i\cup\{i\}} \left(p_{j,i} f(\mathbf{w}_{j,t}, D_i) + \tau p_{j,i}\log p_{j,i}\right)\\ \text{s.t.} \quad & \sum_{j=1}^K p_{j,i} = 1,\\ & p_{j,i} = 0,\ \forall j \notin \mathcal{N}_i \cup \{i\},\\ & p_{j,i} \geq 0, \qquad (11)\end{aligned}$$

where $f(\mathbf{w}_{j,t}, D_i)$ measures the evaluation loss of neighbor model $\mathbf{w}_j^t$ on local validation dataset. This cross validation design provides a common evaluation of distributed model for heterogeneity measurement. $\tau > 0$ controls the smoothness of the learned consensus weights. The entropy regularization $\tau \sum_{j\in\mathcal{N}_i\cup\{i\}} p_{j,i}\log p_{j,i}$ avoids unstable hard selection of a single neighbor and preserves collaborative consensus. Based on the standard first-order optimality conditions [10], proposed entropy-regularized linear objective is strictly convex and yields a Gibbs-form closed-form solution as

$$p_{j,i} = \frac{\exp\left(-\frac{f_i(\mathbf{w}_{j,t}, D_i)}{\tau}\right)}{\sum\limits_{k\in\mathcal{N}_i\cup\{i\}} \exp\left(-\frac{f_i(\mathbf{w}_{k,t}, D_i)}{\tau}\right)}, \quad j \in \mathcal{N}_i \cup \{i\}. \quad (12)$$

In this way, CD-DSL forms a decentralized learning framework is suitable for heterogeneous input in non-convex cases.

D. Algorithm Implementation

Algorithm 1 presents the implementation of CD-DSL. To reduce communication and computation overhead in peer-to-peer wireless networks, CD-DSL uses an asynchronous update schedule with T communication rounds and E local epochs per round. In each round, device i receives model parameters and auxiliary variables from $\mathcal{N}_i \cup \{i\}$, evaluates neighboring models locally, and updates the mixing matrix $\mathbf{P}$ via (12). During local training, each device maintains its local and neighbor best parameters using (7) and (8), updates the model via (10), and transmits the updated variables to its neighbors. This asynchronous mechanism reduces synchronization overhead while maintaining decentralized collaboration.

Algorithm 1 CD-DSL

Initialization:

Initialize $\overline{\mathbf{w}}_t, \mathbf{w}_{i,t}^l, \mathbf{w}_t^n, \mathbf{z}_t$, t, e, i and T, E, given datasets $\{D_i\}$;

for each communication round $t = 1 : T$ **do**
  at edge devices: $i = 1, \ldots, K$
    receive $\{\mathbf{w}_{j,t}, \mathbf{z}_{i,t}\}$ from neighbors in $\mathcal{N}_i \cup \{i\}$ and calculate $\widetilde{\mathbf{w}}_{j,t}$;
    evaluate neighbors' models $f(\mathbf{w}_{j,t}; D_i)$ via local dataset;
    update $\mathbf{P}$ via (12) for consensus and performance based aggregation;
  for each epoch $e = 1 : E$ **do**
    update local and neighbor best via (7) and (8);
    update local model via (10);
    transmit $\{\mathbf{w}_{i,t+1}, \mathbf{z}_{i,t+1}\}$ to neighbors;
  end for
end for

III. Theoretical Analysis

In this section, we analyze convergence behavior of CD-DSL. We start from an ergodic convergence analysis of global model $\overline{\mathbf{w}}$, followed with a study of a non-ergodic convergence neighborhood for all participants $\mathbf{w}_i$. Noting that the proposed consensus mechanism corrects the bias introduced by column-stochastic mixing, the weighted parameters $\widetilde{\mathbf{w}}$ represents a linear combination of $\mathbf{w}$. Thus, the convergence of $\mathbf{w}$ can be derived by $\widetilde{\mathbf{w}}$ with a bounded vector $\mathbf{z}$, which has been proven in [22]. As such, we discuss the convergence of $\widetilde{\mathbf{w}}$ next.

A. Assumptions

Following a similar approach in [13], we adopt the common assumptions apart from Assumption 1 as follows.

Assumption 2. *(Bounded dissimilarity): The deviation between local $\widetilde{\mathbf{w}}_i$ and global $\overline{\mathbf{w}}$ is bounded by a constant σ [9]*

$$\frac{1}{K}\sum_{i=1}^{K}\|\nabla f_i(\widetilde{\mathbf{w}}_i)-\nabla f(\overline{\mathbf{w}})\|^2\leq\sigma^2. \tag{13}$$

Such a bounded dissimilarity assumption benefits in the consensus constraint of data heterogeneity.

Assumption 3. *(L-smoothness): The loss function of local models is differentiable. $\forall i=1,\ldots,K,\exists\{L_i\}_1^K$ so that $\forall\mathbf{w}$, the gradient $\nabla f_i(\mathbf{w}_{i,t})$ bounded by constant L_i that is,*

$$\|\nabla f_i(\widetilde{\mathbf{w}}_{i,t+1})-\nabla f_i(\widetilde{\mathbf{w}}_{i,t})\|\leq L_i\|\widetilde{\mathbf{w}}_{i,t+1}-\widetilde{\mathbf{w}}_{i,t}\|, \tag{14}$$

where L is so called Lipschitz constant [23]. We define the $L_{max}=\max\{L_i\}_i^K$ for the non-ergodic convergence analysis of $\mathbf{f}(\mathbf{X},\mathbf{D})$.

Assumption 4. *(Bounded gradient): The approximated gradient $\nabla f_i\left(\mathbf{w}_{i,t}\right)$ is bounded by [24]*

$$\left\|\nabla f_i\left(\mathbf{w}_{i,t}\right)\right\|^2\leq\phi^2, \tag{15}$$

Assumption 5. *(Bounded local update) For the CD-DSL dynamics [13], [23], there exists a nonnegative sequence $\{\nu_t\}$ such that*

$$\|\tilde{\mathbf{w}}_{i,t}-\bar{\mathbf{w}}_{t-1}\|\leq\nu_t,\ \ \|\mathbf{v}_{i,t}\|=\|\tilde{\mathbf{w}}_{i,t}-\tilde{\mathbf{w}}_{i,t-1}\|\leq\nu_t,\ \forall i,t. \tag{16}$$

Moreover, ν_t is controlled by the local gradient and swarm hyperparameters through

$$\nu_t\leq\alpha_t\phi+|c_0-c_1-c_2|\,\bar{v}_t,\quad \bar{v}_t\triangleq\max_{i=1,\ldots,K}\|\mathbf{v}_{i,t}\|. \tag{17}$$

B. Convergence and Consensus Analysis

For the current round, we define the common model as $\overline{\mathbf{w}}_t\triangleq\frac{1}{K}\sum_{i=1}^{K}\widetilde{\mathbf{w}}_{i,t}$,, the global objective as $F(\overline{\mathbf{w}})\triangleq\frac{1}{K}\sum_{i=1}^{K}f_i(\mathbf{w})$ for convergence analysis and the disagreement $\mathbf{E}_t\triangleq\widetilde{\mathbf{W}}_t\left(\mathbf{I}-\frac{1}{K}\mathbf{1}\mathbf{1}^\top\right)$, consensus error $e_t^2\triangleq\frac{1}{K}\|\mathbf{E}_t\|_F^2=\frac{1}{K}\sum_{i=1}^{K}\|\widetilde{\mathbf{w}}_{i,t}-\overline{\mathbf{w}}_t\|^2$ for consistency analysis.

We first design the common model updates. For the column stochastic mixing matrix $\mathbf{P},0\leq p_{j,i}\leq 1,\forall i,j$, letting local models share the same learning rate α_t, we have:

$$\begin{aligned}\overline{\mathbf{w}}_{t+1}-\overline{\mathbf{w}}_t&=\overline{\boldsymbol{\xi}}_t-\alpha_t\overline{\mathbf{g}}_t\\ \overline{\mathbf{g}}_t&\triangleq\frac{1}{K}\sum_{i=1}^{K}\nabla f_i(\widetilde{\mathbf{w}}_{i,t})\\ \overline{\xi}_t&\triangleq\frac{1}{K}\sum_{i=1}^{K}\xi_{i,t},\end{aligned} \tag{18}$$

where the swarm-induced perturbation term of device i holds:

$$\boldsymbol{\xi}_{i,t}=c_0\mathbf{v}_{i,t}+c_1b_{i,t}\left(\widetilde{\mathbf{w}}_{i,t-1}-\widetilde{\mathbf{w}}_{i,t}\right)+c_2\left(\textstyle\sum_{j=1}^{K}p_{j,i}^t\widetilde{\mathbf{w}}_{j,t}-\widetilde{\mathbf{w}}_{i,t}\right), \tag{19}$$

with $b_{i,t}=\mathbb{I}(f_{i,t-1}>f_{i,t})\in\{0,1\}$ and $\mathbf{v}_{i,t}=\widetilde{\mathbf{w}}_{i,t}-\widetilde{\mathbf{w}}_{i,t-1}$.

Next, we derive the bound of swarm perturbation term.

Lemma 1. *Under Assumption 5, the swarm perturbation matrix $\Xi_t=[\boldsymbol{\xi}_{1,t},\ldots,\boldsymbol{\xi}_{K,t}]$ satisfies*

$$\begin{aligned}&\frac{1}{K}\left\|\Xi_t\left(I-\frac{1}{K}\mathbf{1}\mathbf{1}^\top\right)\right\|_F^2\leq\beta_t^2,\\ &\beta_t\triangleq\left(|c_0|+|c_1|+2|c_2|\right)\overline{\nu}_t.\end{aligned} \tag{20}$$

Proof. Computing norms on both sides of (19) and leveraging the triangle inequality, we have

$$\begin{aligned}\|\boldsymbol{\xi}_{i,t}\|\leq&|c_0|\,\|\mathbf{v}_{i,t}\|+|c_1|\,b_{i,t}\|\tilde{\mathbf{w}}_{i,t-1}-\tilde{\mathbf{w}}_{i,t}\|\\ &+|c_2|\left\|\sum_{j=1}^{K}p_{j,i}^t\tilde{\mathbf{w}}_{j,t}-\tilde{\mathbf{w}}_{i,t}\right\|.\end{aligned} \tag{21}$$

Since $b_{i,t}\leq 1$ and $\tilde{\mathbf{w}}_{i,t-1}-\tilde{\mathbf{w}}_{i,t}=-\mathbf{v}_{i,t}$, we have

$$\|\boldsymbol{\xi}_{i,t}\|\leq(|c_0|+|c_1|)\|\mathbf{v}_{i,t}\|+|c_2|\left\|\textstyle\sum_{j=1}^{K}p_{j,i}^t\tilde{\mathbf{w}}_{j,t}-\tilde{\mathbf{w}}_{i,t}\right\|. \tag{22}$$

Next, we derive the bound of the aggregation mismatch term, by adding and subtracting the common model $\bar{\mathbf{w}}_t$,

$$\begin{aligned}&\left\|\sum_{j=1}^{K}p_{j,i}^t\tilde{\mathbf{w}}_{j,t}-\tilde{\mathbf{w}}_{i,t}\right\|=\left\|\sum_{j=1}^{K}p_{j,i}^t(\tilde{\mathbf{w}}_{j,t}-\bar{\mathbf{w}}_t)-(\tilde{\mathbf{w}}_{i,t}-\bar{\mathbf{w}}_t)\right\|\\ &\leq\sum_{j=1}^{K}p_{j,i}^t\|\tilde{\mathbf{w}}_{j,t}-\bar{\mathbf{w}}_t\|+\|\tilde{\mathbf{w}}_{i,t}-\bar{\mathbf{w}}_t\|.\end{aligned} \tag{23}$$

Since P_t is column stochastic as $\sum_{j=1}^{K}p_{j,i}^t=1,p_{j,i}^t\geq 0$, and Assumption 5 ensures $\|\tilde{\mathbf{w}}_{j,t}-\bar{\mathbf{w}}_t\|\leq\overline{\nu}_t$ and $\|\tilde{\mathbf{w}}_{i,t}-\bar{\mathbf{w}}_t\|\leq\overline{\nu}_t$, according to (23), we have:

$$\left\|\sum_{j=1}^{K}p_{j,i}^t\tilde{\mathbf{w}}_{j,t}-\tilde{\mathbf{w}}_{i,t}\right\|\leq 2\nu_t. \tag{24}$$

Then, substituting (24) and leveraging Assumption 5 into (22), it yields:

$$\|\boldsymbol{\xi}_{i,t}\|\leq\left(|c_0|+|c_1|+2|c_2|\right)\nu_t\triangleq\beta_t. \tag{25}$$

Finally, given (25) and $\left\|I-\frac{1}{K}\mathbf{1}\mathbf{1}^\top\right\|_2\leq 1$, we obtain

$$\begin{aligned}\frac{1}{K}\left\|\Xi_t\left(I-\frac{1}{K}\mathbf{1}\mathbf{1}^\top\right)\right\|_F^2&\leq\frac{1}{K}\|\Xi_t\|_F^2=\frac{1}{K}\sum_{i=1}^{K}\|\boldsymbol{\xi}_{i,t}\|^2\\ &\leq\frac{1}{K}\sum_{i=1}^{K}\beta_t^2=\beta_t^2.\end{aligned} \tag{26}$$

This completes the proof. □

Next, we bound the consensus error and heterogeneous bias.

Lemma 2. *Consider the distributed data input heterogeneity, the bias of common and local gradient is bounded by*

$$\|\overline{\mathbf{g}}_t-\nabla F(\overline{\mathbf{w}}_t)\|^2\leq 2L_{\max}^2e_t^2+\sigma^2. \tag{27}$$

Proof. By the triangle inequality, we have

$$\begin{aligned}\overline{\mathbf{g}}_t-\nabla F(\overline{\mathbf{w}}_t)=&\frac{1}{K}\sum_{i=1}^{K}\left(\nabla f_i(\widetilde{\mathbf{w}}_{i,t})-\nabla f_i(\overline{\mathbf{w}}_t)\right)\\ &+\frac{1}{K}\sum_{i=1}^{K}\left(\nabla f_i(\overline{\mathbf{w}}_t)-\nabla F(\overline{\mathbf{w}}_t)\right),\end{aligned} \tag{28}$$

Given Assumption 2 and applying Jensen's inequality, it has

$$\left\| \frac{1}{K} \sum_{i=1}^{K} \left(\nabla f_i(\widetilde{\mathbf{w}}_{i,t}) - \nabla f_i(\overline{\mathbf{w}}_t) \right) \right\|^2 \leq L_{\max}^2 e_t^2. \tag{29}$$

Assumption 2 bounds the heterogeneity term. Combining two terms and applying $(a+b)^2 \leq 2a^2 + 2b^2$, it yields (27). □

Theorem 1 (Ergodic convergence)**.** *Suppose all Assumptions hold. Let the learning rate satisfy*

$$0 < \alpha_t \leq \bar{\alpha} \triangleq \min \left\{ \frac{1}{4L_{\max}}, \frac{1-\lambda}{2\sqrt{2}L_{\max}} \right\}, \tag{30}$$

where $\lambda \triangleq \sup_t |\lambda_2(P_t)| < 1$. *Then, for any* $T \geq 1$, *it holds:*

$$\frac{\sum_{t=0}^{T-1} \alpha_t \mathbb{E}\|\nabla F(\overline{\mathbf{w}}_t)\|^2}{\sum_{t=0}^{T-1} \alpha_t} \leq \frac{4\left(F(\overline{\mathbf{w}}_0) - F(\overline{\mathbf{w}}^\star)\right)}{\sum_{t=0}^{T-1} \alpha_t} + \frac{3\sum_{t=0}^{T-1} \mathbb{E}[e_t^2]/\alpha_t}{8\sum_{t=0}^{T-1} \alpha_t} + 3\sigma^2 + \frac{6\sum_{t=0}^{T-1} \mathbb{E}[\beta_t^2]/\alpha_t}{\sum_{t=0}^{T-1} \alpha_t}. \tag{31}$$

Proof. Given $L_{\max}$-smoothness of F, it has

$$\begin{aligned} F(\overline{\mathbf{w}}_{t+1}) \leq & F(\overline{\mathbf{w}}_t) + \frac{L_{\max}}{2}\|\overline{\mathbf{w}}_{t+1} - \overline{\mathbf{w}}_t\|^2 \\ & + \langle \nabla F(\overline{\mathbf{w}}_t), \overline{\mathbf{w}}_{t+1} - \overline{\mathbf{w}}_t \rangle \\ \leq & F(\overline{\mathbf{w}}_t) + \frac{L_{\max}}{2}\|\overline{\boldsymbol{\xi}}_t - \alpha_t \overline{\mathbf{g}}_t\|^2 \\ & + \left\langle \nabla F(\overline{\mathbf{w}}_t), \overline{\boldsymbol{\xi}}_t \right\rangle - \left\langle \nabla F(\overline{\mathbf{w}}_t), \alpha_t \overline{\mathbf{g}}_t \right\rangle. \end{aligned} \tag{32}$$

Substituting Lemma 2, Lemma. 1 and Young's inequality into (32), it produces:

$$\begin{aligned} & F(\overline{\mathbf{w}}_{t+1}) \leq F(\overline{\mathbf{w}}_t) - \alpha_t \|\nabla F(\overline{\mathbf{w}}_t)\|^2 \\ & + \frac{L_{\max}}{2}\|\overline{\boldsymbol{\xi}}_t - \alpha_t \overline{\mathbf{g}}_t\|^2 + \left\langle \nabla F(\overline{\mathbf{w}}_t), \overline{\boldsymbol{\xi}}_t - \alpha_t \overline{\mathbf{g}}_t + \alpha_t \right\rangle \\ \leq & F(\overline{\mathbf{w}}_t) - \alpha_t \|\nabla F(\overline{\mathbf{w}}_t)\|^2 + \frac{\alpha_t}{2}\|\nabla F(\overline{\mathbf{w}}_t)\|^2 \\ & + \frac{1}{2\alpha_t}\|\overline{\boldsymbol{\xi}}_t - \alpha_t \overline{\mathbf{g}}_t + \alpha_t \nabla F(\overline{\mathbf{w}}_t)\|^2 + L_{max}\alpha_t^2 \|\nabla F(\overline{\mathbf{w}}_t)\|^2 \\ & + L_{max}\|\overline{\boldsymbol{\xi}}_t - \alpha_t \overline{\mathbf{g}}_t + \alpha_t \nabla F(\overline{\mathbf{w}}_t)\|^2. \\ \leq & F(\overline{\mathbf{w}}_t) + \left(L_{max}\alpha_t^2 - \frac{\alpha_t}{2} \right) \|\nabla F(\overline{\mathbf{w}}_t)\|^2 \\ & + \left(L_{max} + \frac{1}{2\alpha_t} \right) \left(\alpha_t^2 \| - \overline{\mathbf{g}}_t + \nabla F(\overline{\mathbf{w}}_t)\|^2 + 2\|\overline{\boldsymbol{\xi}}_t\|^2 \right). \\ \leq & F(\overline{\mathbf{w}}_t) + \left(L_{max}\alpha_t^2 - \frac{\alpha_t}{2} \right) \|\nabla F(\overline{\mathbf{w}}_t)\|^2 \\ & + \left(L_{max} + \frac{1}{2\alpha_t} \right) \left(\alpha_t^2 \left(2L_{\max}^2 e_t^2 + \sigma^2 \right) + 2\beta_t^2 \right). \end{aligned} \tag{33}$$

To ensure convergence, α_t should satisfy $L_{max}\alpha_t^2 - \frac{\alpha_t}{2} < 0, 0 < \alpha_t$, $\alpha_t < \frac{1}{2L_{max}}$. For a proper design of learning rate, we set $\alpha_t \leq \frac{1}{4L_{max}}$ to minimize the $L_{max}\alpha_t^2 - \frac{\alpha_t}{2}$, in order to make (32) satisfy:

$$\begin{aligned} F(\overline{\mathbf{w}}_{t+1}) \leq & F(\overline{\mathbf{w}}_t) - \frac{\alpha_t}{4}\|\nabla F(\overline{\mathbf{w}}_t)\|^2 \\ & + \frac{3e_t^2}{32\alpha_t} + \frac{3\alpha_t\sigma^2}{4} + \frac{3}{2\alpha_t}\beta_t^2. \end{aligned} \tag{34}$$

As the model eventually converge at $\overline{\mathbf{w}}^\star$ when $t \to \infty$, the accumulated expectation of global model gradients from $t = 0$ to $T-1$ holds

$$\begin{aligned} & \lim_{T\to\infty} \sum_{t=0}^{T} \frac{\alpha_t}{4} \mathbb{E}\left[\|\nabla F(\overline{\mathbf{w}}_t)\|^2\right] \\ \leq & \left(F(\overline{\mathbf{w}}_0) - F(\overline{\mathbf{w}}^\star)\right) + \frac{3\sum_{t=0}^{T-1} \mathbb{E}[e_t^2]/\alpha_t}{32} \\ & + \frac{3\sigma^2 \sum_{t=0}^{T-1} \alpha_t}{4} + \frac{3\sum_{t=0}^{T-1} \mathbb{E}[\beta_t^2]/\alpha_t}{2}, \end{aligned} \tag{35}$$

which thus yields (31). □

Theorem 2 (Bounded consensus bias)**.** *Suppose all Assumptions hold. Then the consensus error satisfies*

$$e_{t+1} \leq \left(\lambda + \sqrt{2}L_{\max}\alpha_t \right) e_t + \sqrt{2}\sigma\alpha_t + \beta_t, \tag{36}$$

where β_t *is the effective swarm-induced perturbation bound derived from Assumption 5, as characterized in Lemma 1. Consequently, if*

$$0 < \alpha_t \leq \frac{1-\lambda}{2\sqrt{2}L_{\max}}, \tag{37}$$

and define the

$$\rho_t \triangleq \lambda + \sqrt{2}L_{\max}\alpha_t < 1, \tag{38}$$

then

$$e_t \leq \left(\prod_{s=0}^{t-1} \rho_s \right) e_0 + \sum_{s=0}^{t-1} \left(\prod_{\ell=s+1}^{t-1} \rho_\ell \right) \left(\sqrt{2}\sigma\alpha_s + \beta_s \right). \tag{39}$$

Proof. From (10), we have

$$\widetilde{\mathbf{W}}_{t+1} = \widetilde{\mathbf{W}}_t \mathbf{P}_t - \alpha_t \mathbf{G}_t + \boldsymbol{\Xi}_t, \tag{40}$$

where $\mathbf{G}_t \triangleq \left[\nabla f_1(\widetilde{\mathbf{w}}_{1,t}), \ldots, \nabla f_K(\widetilde{\mathbf{w}}_{K,t})\right], \boldsymbol{\Xi}_t \triangleq \left[\boldsymbol{\xi}_{1,t}, \ldots, \boldsymbol{\xi}_{K,t}\right]$. Multiplying both sides of (40) by $\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top$, we obtain:

$$\begin{aligned} \mathbf{E}_{t+1} = & \widetilde{\mathbf{W}}_t \left(\mathbf{P}_t - \frac{1}{K}\mathbf{1}\mathbf{1}^\top \right) - \alpha_t \mathbf{G}_t \left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top \right) \\ & + \boldsymbol{\Xi}_t \left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top \right). \end{aligned} \tag{41}$$

Using the spectral property of $\mathbf{P}_t$, $\left\|\mathbf{P}_t - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right\| \leq \lambda$, we have:

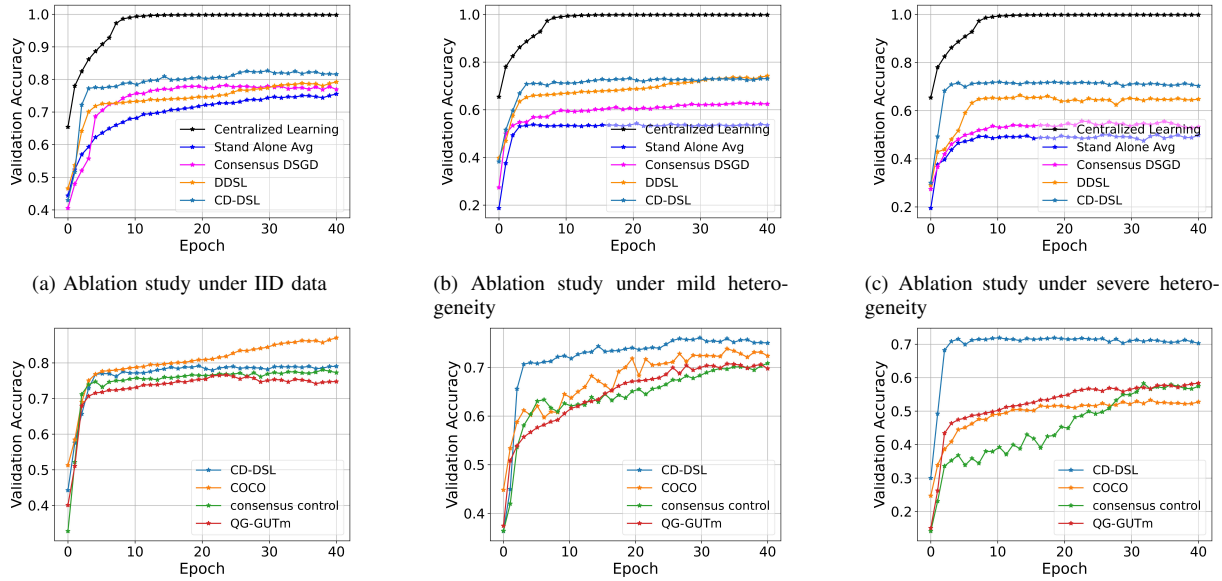
$$\begin{aligned} e_{t+1} \leq & \lambda e_t + \alpha_t \left(\frac{1}{K} \left\| \mathbf{G}_t \left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top \right) \right\|_F^2 \right)^{1/2} \\ & + \left(\frac{1}{K} \left\| \boldsymbol{\Xi}_t \left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top \right) \right\|_F^2 \right)^{1/2}. \end{aligned} \tag{42}$$

By leveraging Lemma 2 and Lemma 1, it leads

$$\begin{aligned} \frac{1}{K} \left\| \mathbf{G}_t \left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top \right) \right\|_F^2 \leq & 2L_{\max}^2 e_t^2 + 2\sigma^2 \\ \frac{1}{K} \left\| \boldsymbol{\Xi}_t \left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top \right) \right\|_F^2 \leq & \beta_t^2. \end{aligned} \tag{43}$$

Finally, substituting the above two bounds into (36), it yields (39). □

Corollary 1 (Non-ergodic convergence)**.** *Let* $\alpha_t = \alpha_0\sqrt{\frac{K}{T}}, \beta_t \leq \overline{\beta}\alpha_t$, *[19] with* α_0 *satisfying* (30). *The participant of* K*-clients CD-DSL achieves*



(a) Ablation study under IID data

(b) Ablation study under mild heterogeneity

(c) Ablation study under severe heterogeneity

(d) Comparison under IID data

(e) Comparison under mild heterogeneity

(f) Comparison under severe heterogeneity

Fig. 1: Performance evaluation of the proposed CD-DSL. The first row reports the ablation results of CD-DSL and its key components under different data distributions. The second row compares CD-DSL with representative decentralized baselines.

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(\widetilde{\mathbf{w}}_{i,t})\|^2 = \mathcal{O}\left(\frac{1}{\alpha_0\sqrt{KT}}\right) + \mathcal{O}(\sigma^2) + \mathcal{O}\left(\frac{K\alpha_0^2(\sigma+\overline{\beta})^2}{T}\right), \tag{44}$$

and the consensus bias obeys

$$\limsup_{t\to\infty} e_t \leq \frac{K\alpha_0^2(\sqrt{2}\sigma+\overline{\beta})}{T(1-\lambda-\sqrt{2}L_{\max}\alpha_0\sqrt{K/T})}. \tag{45}$$

$$e_t = \mathcal{O}\left(\frac{1}{\alpha_0\sqrt{KT}}\right) + \mathcal{O}(\rho^t), \qquad \rho < 1. \tag{46}$$

Remark. In summary, the result of Theorem 1 reveals that the common model $\overline{\mathbf{w}}_t$ will converge to $\overline{\mathbf{w}}^\star$. Theorem 2 and Corollary 1 together show that local models $\{\mathbf{w}_i\}$ will converge to an $\mathcal{O}(\sum_{s=0}^{t-1}\left(\prod_{\ell=s+1}^{t-1}\rho_\ell\right)(\sqrt{2}\sigma\alpha_s+\beta_s))$-neighborhood of the unique solution $\overline{\mathbf{w}}^\star$.

IV. Simulation Results

We test different decentralized learning methods via numerical experiments for image classification on CIFAR10 with various settings [25]. We focus on the non-convex parameter optimization tasks by evaluating the cross-entropy loss.

A. Experimental Settings

We design a decentralized system in Ubuntu 20.04.6 operating system with 4 Nvidia V100-SXM2 GPUs to simulate the collaborative learning across $K = 50$ edge devices [1]. Each contains an individual dataset $|D_i| = 1000$ and a validation dataset $|D_i^v| = 128$. Distributed data is deployed on a strongly connected decentralized edge network topology with given connection rate 0.7. The heterogeneous data generation relies on the Dirichlet distribution with concentration parameter α. The sampled training and validation datasets share different label distribution across distributed devices. We design a IID case and two non-IID cases: mild heterogeneity where each tasks with $\alpha = 0.5$ and severe heterogeneity where 20 sets sampled by $\alpha = 0.1$, 15 sets by $\alpha = 0.5$, 10 sets by $\alpha = 1$ and 5 sets by $\alpha = 10$. The ResNet18 model is employed for CIFAR10 classification.

For the hyperparameter settings, we apply the SGD optimizer with attenuated learning rate $\alpha_{init} = 0.01, \gamma = 0.5$. The training process involves 40 communication rounds on CIFAR10, each round includes 6 epochs. The batch size is 32 for stable training. The hyperparameters of PSO updates follow $c_0 \sim U\left(0, 0.05(0.98)^t\right)$ for controlled exploration and $c_1, c_2 \sim \mathcal{N}\left(0, 0.15\right)$ for balanced exploitation. The consensus weights in joint optimization denotes $\tau = 0.09$. Comparative experiments about COCO [4], consensus control [9], and QG-GUTm [19] are designed with optimal settings under given wireless topology. Throughout training, we evaluate and record the validation accuracy of the global model on an unseen validation dataset.

B. Performance Evaluation and Comparison

Fig. 1 reports the validation accuracy of CD-DSL compared with baseline methods under the IID case, mild heterogeneity, and the more challenging severe heterogeneity.

The ablation results in Figs. 1a–1c show that the proposed CD-DSL consistently outperforms its counterpart versions such as consensus optimization and DDSL alone. Under the IID setting in Fig. 1a, all decentralized variants can achieve stable convergence, while CD-DSL reaches higher accuracy at

[1]The code is available at https://github.com/zhuoyu-3/CD-DSL.

faster convergence speed than consensus-based DSGD. Such advantages become obvious as data heterogeneity increases. As shown in Fig. 1b, CD-DSL maintains stable convergence and higher final accuracy, whereas DDSL saturates earlier. Such performance enhancement provided by CD-DSL beyond the benchmarks further increases under severe heterogeneity in Fig. 1c, where CD-DSL remains working well but the ablated methods suffer from slower convergence and lower accuracy. These results indicate that consensus optimization improves model alignment, while DSL-enabled swarm updates enhance exploration and exploitation from non-IID data.

The comparison results in Figs. 1d–1f further validate the robustness of CD-DSL beyond the other decentralized baselines. Although COCO achieves competitive performance under the IID setting in Fig. 1d, CD-DSL is more robust against non-IID issues as the data distribution becomes more heterogeneous. In the mild heterogeneity case, Fig. 1e shows that CD-DSL converges faster and achieves higher validation accuracy than COCO, consensus control, and QG-GUTm. This advantage becomes more evident in the severe heterogeneity case shown in Fig. 1f, where CD-DSL maintains a large performance margin over all baselines in terms of both convergence speed and converged accuracy. Overall, the results demonstrate that the integration of push-sum consensus, performance-aware mixing, and DSL-enabled swarm updates allows CD-DSL to achieve stronger model consistency and robustness in decentralized learning with heterogeneous data.

V. Conclusion

This paper develops a consensus-based decentralized distributed swarm learning (CD-DSL) framework for edge intelligence with heterogeneous big data. By integrating push-sum consensus, adaptive neighbor mixing, and bio-inspired model updates, the proposed CD-DSL enables decentralized learning over directed wireless topologies. Theoretical analysis shows that CD-DSL achieves consensus and converges to a neighborhood of a stationary point under non-IID data distributions and non-convex objectives. Experimental results validate the robustness and efficiency of CD-DSL in heterogeneous decentralized learning scenarios.

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